

Circular Economy Ontology Network (CEON) - Product Module

Metadata

IRI

<http://w3id.org/CEON/ontology/product/>

Title

Circular Economy Ontology Network (CEON) - Product Module

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Version Iri

<http://w3id.org/CEON/ontology/product/0.2/>

Version Info

0.2

Preferred Namespace Prefix

product

Preferred Namespace Uri

<http://w3id.org/CEON/ontology/product/>

Description

The Product module of CEON (Circular Economy Ontology Network).

Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-2, CVN-ResourceType-4, C11-2, C12-1, C13-3, E2-1, T8-3.

Classes

Matter composition^c

IRI

<http://w3id.org/CEON/ontology/product/MatterComposition>

Description

A product composition is the composed information of a product in terms of a specific component, in which such a component is also a matter.

Sub Class Of

[resourceODP:Composition](#)

In Domain Of

[associated with matter](#)^{op}

Restriction

[associated with matter](#)^{op} *exactly* 1 [Matter composition](#)^c

Product model^C

IRI	http://w3id.org/CE0N/ontology/product/Product
Description	Represents the common sense notion of a product, i.e. the abstract notion of a product type or product model, which is a model that is used for manufacturing products.
In Range Of	associated with product model ^{OP} batch of products ^{OP} modelled by ^{OP}
Restriction	hasComposition ^{OP} some Product model ^C hasComposition ^{OP} some Product model ^C

Product composition^C

IRI	http://w3id.org/CE0N/ontology/product/ProductComposition
Description	A product composition is the composed information of a product in terms of a specific component, in which such a component is also a product.
Sub Class Of	resourceODP:Composition
In Domain Of	associated with product model ^{OP}
Restriction	associated with product model ^{OP} <i>exactly</i> 1 Product composition ^C

Product object^C

IRI	http://w3id.org/CE0N/ontology/product/ProductObject
Description	A product object is a physical object put into a market for sale, i.e. corresponding to the notion of an item, which conforms to the abstract idea of a product model. The product can be a thing that grows naturally or produced through some chemical or manufacturing processes.
Sub Class Of	resourceODP:PhysicalObject
In Domain Of	has product component ^{OP} modelled by ^{OP}
In Range Of	has product component ^{OP}
Restriction	modelled by ^{OP} <i>exactly</i> 1 Product object ^C

Object Properties

associated with matter^{op}

IRI	http://w3id.org/CE0N/ontology/product/associatedWithMatter
Description	associatedWithMatter intends to represent the matter to which a matter composition information regards to.
Domain	Matter composition ^c
Range	resourceODP:Matter

associated with product model^{op}

IRI	http://w3id.org/CE0N/ontology/product/associatedWithProductModel
Description	associatedWithProductModel intends to represent the matter to which a product composition information regards to.
Domain	Product composition ^c
Range	Product model ^c

batch of products^{op}

IRI	http://w3id.org/CE0N/ontology/product/batchOfProduct
Description	batchOfProduct intends to represent what is the model product of a batch of objects.
Domain	resourceODP:BatchOfObjects
Range	Product model ^c

has composition^{op}

IRI	http://w3id.org/CE0N/ontology/product/hasComposition
Description	hasComposition intends to represent that a product model or physical object can have composition information.
Domain	resourceODP:PhysicalObject ^c or Product model ^c
Range	resourceODP:Composition

has product component^{op}

IRI	http://w3id.org/CE0N/ontology/product/hasProductComponent
Description	hasProductComponent intends to represent that a product can have other product components.
Domain	Product object ^c
Range	Product object ^c

modelled by^{op}

IRI	http://w3id.org/CE0N/ontology/product/modelledBy
Description	modelledBy intends to represent that product objects follow particular product models.
Domain	Product object ^c
Range	Product model ^c

Datatype Properties

Composition quantity^{dp}

IRI	http://w3id.org/CE0N/ontology/product/compositionQuantity
Description	compositionQuantity intends to represent the quantity value of composed elements of a composition. The quantity value can be the number of a element or the percentage of a element.
Domain	resourceODP:Composition
Range	xsd:double ^c or xsd:positiveInteger ^c

Annotation Properties

contributor^{ap}

IRI	http://purl.org/dc/terms/contributor
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created^{ap}

IRI	http://purl.org/dc/terms/created
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creator^{ap}

IRI	http://purl.org/dc/terms/creator
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description ^{ap}	
IRI	http://purl.org/dc/terms/description
license ^{ap}	
IRI	http://purl.org/dc/terms/license
title ^{ap}	
IRI	http://purl.org/dc/terms/title
preferred namespace prefix ^{ap}	
IRI	http://purl.org/vocab/vann/preferredNamespacePrefix
preferred namespace uri ^{ap}	
IRI	http://purl.org/vocab/vann/preferredNamespaceUri
covers requirements ^{ap}	
IRI	http://www.ontologydesignpatterns.org/schemas/cannotation schema.owl#coversRequirements

Namespaces

: http://w3id.org/CEON/ontology/product/
dcterms http://purl.org/dc/terms/
odp http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#
owl http://www.w3.org/2002/07/owl#
prov http://www.w3.org/ns/prov#
rdf http://www.w3.org/1999/02/22-rdf-syntax-ns#
rdfs http://www.w3.org/2000/01/rdf-schema#
resourceODP http://w3id.org/CEON/ontology/resourceODP/
vann http://purl.org/vocab/vann/
xsd http://www.w3.org/2001/XMLSchema#

Legend

c	Classes
op	Object Properties
dp	Datatype Properties
ap	Annotation Properties